



Fig. 6: Single-ended and differential digital signals

frequency sweep function in your signal generator. It generates a sine wave whose frequency changes over time. This frequency change occurs linearly in Hz/second or logarithmically. Some signal generators even provide sweep sequencing facility, where start, stop, hold frequency and associated time can be defined. Some signal generators also provide a trigger signal synchronously to the sweep to control an oscilloscope that measures the output response of the device.

Digital patterns. A digital pattern consists of multiple synchronised pulse streams that make up 'words' of data that may be 8, 12, 16 or more bits wide. The most common signal formats that also come under complex waveforms category are:

1. Non-return-to-zero
2. Delayed non-return-to-zero
3. Return-to-zero
4. Return-to-one

Only specialised signal generators support these waveforms.

Performance specifications

Now after discussing the waveform-related specifications, given below are some performance-related speci-

fications of a signal generator. These specifications have to be checked carefully to select the best signal generator in the range.

Memory depth. Also known as record length, it determines the maximum number of samples that can be stored. Memory depth plays an important role in signal fidelity at many frequencies because it determines how many points of data can be stored to define a waveform. Particularly in the case of complex waveforms, memory depth is critical to reproducing signal details accurately. High-performance mixed-signal generators offer large memory depth and high sample rates. These instruments can store and reproduce complex waveforms such as pseudo-random bitstreams.

Sample rate. The Nyquist Sampling Theorem states that the sampling frequency must be more than twice the highest spectral frequency component of the generated signal to ensure accurate signal reproduction. For example, to generate a 2MHz sinewave signal, it is necessary to produce sample points at a frequency of more than 4 megasamples per second (MSA/s). Al-

though the theorem is usually cited for digital oscilloscopes, it works equally well for signal generators. The stored waveform must have enough points so that the signal can be reproduced reliably. Go for a signal generator with the highest available sample rate specification.

Bandwidth. The analogue bandwidth of the output circuitry should be large enough to support the highest frequency signal that the signal generator can produce. This bandwidth is independent of the sample rate. If there is not enough bandwidth, the signal characteristics may degrade.

Horizontal and vertical resolution. Horizontal resolution is the smallest time increment that can be used to create a waveform. It is actually inversely proportional to the sampling frequency. By this definition, the timing resolution of a signal generator with maximum clock rate of 20 MHz would be 50 nanoseconds. On the other hand, vertical resolution of the DAC defines the amplitude accuracy and distortion of the reproduced waveform. A DAC with inadequate resolution contributes to quantisation errors, causing imperfect waveform generation.

Output channels. Many applications require more than one output channel from the signal generator. Some signal generators can deliver up to four independent channels with full bandwidth. Others offer up to two analogue outputs, supplemented by some high-speed digital outputs for mixed-signal testing. Check in advance your requirement of the number of outputs because this will considerably add to the overall price. ●

The author is a technical editor at EFY



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